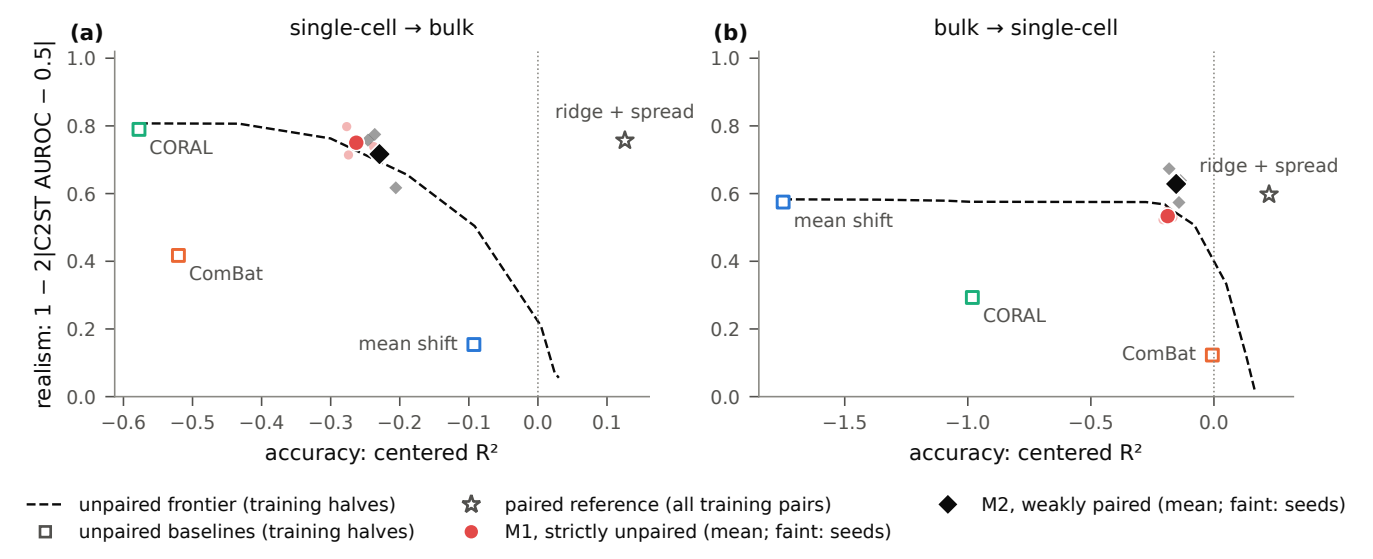
(a)
single-cell → bulk
realism: 1 − 2|C2ST AUROC − 0.5|
accuracy: centered R²
CORAL
ComBat
mean shift
ridge + spread
(b)
bulk → single-cell
mean shift
CORAL
ComBat
ridge + spread
accuracy: centered R²
unpaired frontier (training halves)
unpaired baselines (training halves)
paired reference (all training pairs)
M1, strictly unpaired (mean; faint: seeds)
M2, weakly paired (mean; faint: seeds)